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Department of Mathematics
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Math 208 Major Exam I Term 232

Original exam date: 20 February 2024. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. The general solution of the differential equation $xy'' = y'$ is given by

(a) $y = Ax^2 + Bx$

(b) $y = Ax^3 + B$

(c) $y = Ax^3 + Bx$

(d) $y = Ax^2 + B$

(e) $y = Ax^4 + B$

2. The general solution of the linear differential equation $xy' = 3y + x^4 \cos x$ is given by

(a) $y = x^3(\cos x + c)$

(b) $y = x^2(\cos x + c)$

(c) $y = x^4(\sin x + c)$

(d) $y = x^2(\sin x + c)$

(e) $y = x^3(\sin x + c)$

3. By making a suitable substitution, the differential equation $(x^2 - y^2)y' = 2xy$ can be transformed into a separable differential equation

(a) $\dfrac{dy}{y} - \dfrac{2v}{v^2+1}\,dv = 0$

(b) $\dfrac{dy}{y} + \dfrac{v}{v^2+1}\,dv = 0$

(c) $\dfrac{dy}{y} + \dfrac{3v}{v^2+1}\,dv = 0$

(d) $\dfrac{dy}{y} - \dfrac{v}{v^2+1}\,dv = 0$

(e) $\dfrac{dy}{y} + \dfrac{2v}{v^2+1}\,dv = 0$

4. If a particle is moving in a straight line with acceleration $a(t) = \dfrac{1}{(t+1)^3}$, initial position $x(0) = 0$, and an initial velocity $v(0) = 0$, then the position of the particle at $t = \dfrac{1}{2}$ is

(a) $\dfrac{1}{12}$

(b) $\dfrac{1}{14}$

(c) $\dfrac{2}{3}$

(d) $\dfrac{1}{2}$

(e) $\dfrac{3}{2}$

5. Let $\mathbf{a} = 2\mathbf{i} - \mathbf{j}$, $\mathbf{b} = \mathbf{j} - 3\mathbf{k}$. Then $|3\mathbf{a} - 4\mathbf{b}| =$

(a) $\sqrt{231}$

(b) $\sqrt{226}$

(c) $\sqrt{229}$

(d) $\sqrt{224}$

(e) $\sqrt{227}$

6. If $y = \dfrac{1}{4}x^5 + bx^{-3}$ is a solution of the initial-value problem

$$x\frac{dy}{dx} + 3y = 2x^5, \quad y(2) = 1,$$

then $b =$

(a) -56

(b) 53

(c) -57

(d) -55

(e) 54

7. A constant K that makes the differential equation

$$\left(x^3 + \frac{y}{x}\right) dx + (y^2 + K \ln x)\, dy = 0$$

exact is

(a) 4

(b) 1

(c) 5

(d) 2

(e) 3

8. By making a suitable substitution, the differential equation $2xy' + y^3 e^{-2x} = 2xy$ can be transformed into a linear differential equation

(a) $u' + 2u = \dfrac{e^{-2x}}{x}$

(b) $u' + 2u = \dfrac{e^{-2x}}{3x}$

(c) $u' - 2u = \dfrac{e^{-2x}}{x}$

(d) $u' - 3u = \dfrac{e^{-2x}}{x}$

(e) $u' + 3u = \dfrac{e^{-2x}}{x}$

9. The value of k for which the vectors of $\mathbb{R}^3$

$$\mathbf{u} = (1, -1, 2), \quad \mathbf{v} = (3k, 0, k + 1), \quad \mathbf{w} = (1, -2, 2)$$

are linearly independent are

(a) $k \neq \dfrac{1}{6}$

(b) $k \neq 0$

(c) $k \neq 1$

(d) $k \neq \dfrac{1}{4}$

(e) $k \neq \dfrac{1}{5}$

10. The order of the differential equation

$$y^{(4)} + x^2 y^{(3)} + x(y')^6 = \sin x$$

is equal to

(a) 3

(b) 2

(c) 4

(d) 6

(e) 7

11. Let $\mathbf{u} = (4, 1)$, $\mathbf{v} = (-2, -1)$, $\mathbf{w} = (2, -2)$ be vectors in $\mathbb{R}^2$. If $\mathbf{w} = a\mathbf{u} + b\mathbf{v}$, then $a + b =$

(a) 5

(b) 4

(c) 8

(d) 6

(e) 7

12. The solution of the system

$$\begin{aligned} x_1 + 3x_2 + 3x_3 + 3x_4 &= 0 \\ 2x_1 + 7x_2 + 5x_3 - x_4 &= 0 \\ 2x_1 + 7x_2 + 4x_3 - 4x_4 &= 0 \end{aligned}$$

is the set of all scalar multiples of a vector $\mathbf{u}$ where $\mathbf{u} =$

(a) $(-6, -4, -3, 1)$

(b) $(-6, 4, -3, -1)$

(c) $(-6, 4, 3, 1)$

(d) $(6, 4, -3, 1)$

(e) $(-6, 4, -3, 1)$

13. The general solution of the differential equation $(x^2 + 1)(\tan y)\, y' = x$ is given by

(a) $y(x) = \tan^{-1}\left(c\sqrt{1 + x^2}\right)$

(b) $y(x) = \sec^{-1}\left(c\sqrt{2 + x^2}\right)$

(c) $y(x) = \tan^{-1}\left(c\sqrt{2 + x^2}\right)$

(d) $y(x) = \sec^{-1}\left(c\sqrt{x^2 - 1}\right)$

(e) $y(x) = \sec^{-1}\left(c\sqrt{1 + x^2}\right)$

14. A general solution of the differential equation $y' + \dfrac{1}{3x}\, y = 4$ is

(a) $y = x + \dfrac{c}{\sqrt[3]{x}}$

(b) $y = 3x + \dfrac{c}{\sqrt{x}}$

(c) $y = 3x + \dfrac{c}{\sqrt[3]{x}}$

(d) $y = x + \dfrac{c}{\sqrt{x}}$

(e) $y = 2x + \dfrac{c}{\sqrt{x}}$

15. Which one of the following differential equations is linear?

(a) $y' + x \sin y = e^x$

(b) $(y')^2 + xy = x^2$

(c) $y' + xy^2 = x^3$

(d) $yy' + (\ln x)\, y = e^x$

(e) $e^x y' + (\ln x)\, y = \tan x$

16. Which one of the following statements is true about the subset V of $\mathbb{R}^3$ defined by

$$V = \{(x, y, z) : x + y + z = 0\}?$$

(a) V is not closed under addition

(b) V is not closed under multiplication by scalars

(c) V is not closed under addition but closed under multiplication by scalars

(d) V is a subspace of $\mathbb{R}^3$

(e) V is closed under addition but not closed under multiplication by scalars

17. If $y = y(x)$ is the solution of the initial-value problem

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}, \quad y(0) = 0,$$

then $y\left(\frac{1}{2}\right) =$

(a) $\frac{\pi}{4}$

(b) $-\frac{\pi}{6}$

(c) $-\frac{\pi}{3}$

(d) $\frac{\pi}{6}$

(e) $\frac{\pi}{3}$

18. A pitcher of buttermilk initially at 25°C is to be cooled by setting it on the front porch, where the temperature is 0°. Suppose that the temperature of the buttermilk has dropped to 15° after 20 min. The temperature of the buttermilk will be 9° when $t =$

(a) 25

(b) 28

(c) 20

(d) 30

(e) 40

19. A general solution of the exact differential equation

$$\left(\frac{2x}{y} - \frac{3y^2}{x^4}\right)dx + \left(\frac{2y}{x^3} - \frac{x^2}{y^2} + \frac{1}{\sqrt{y}}\right)dy = 0$$

is

(a) $\frac{x}{y} + \frac{y^2}{x^3} + 4\sqrt{y} = c$

(b) $\frac{x^2}{y} - \frac{x^3}{y^2} + 3\sqrt{y} = c$

(c) $\frac{x^2}{y} + \frac{y^2}{x^3} + 2\sqrt{y} = c$

(d) $\frac{x^2}{y} + \frac{y^2}{x^3} - 3\sqrt{y} = c$

(e) $\frac{y}{x^2} + \frac{y^2}{x^3} + 2\sqrt{y} = c$

20. A general solution of the differential equation $(x+y)\,y' = 1$ is

(a) $y = \ln(x+y+1) + cx$

(b) $y = \ln(x+y-1) + cx$

(c) $y = \ln(x+y+2) + c$

(d) $y = \ln(x+y+1) + c$

(e) $y = \ln(x+y+2) + cx$

Answer Key

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Q	Ans	Brief explanation
1	**(d)**	$p = y'$: $xp' = p$, $p = Cx$, so $y = Ax^2 + B$.
2	**(e)**	$\rho = x^{-3}$: $(x^{-3}y)' = \cos x$, so $y = x^3(\sin x + c)$.
3	**(e)**	$x = vy$: $y\frac{dv}{dy} = -\frac{v^2+1}{2v}$, so $\frac{dy}{y} + \frac{2v}{v^2+1}dv = 0$.
4	**(a)**	$v = \frac{1}{2} - \frac{1}{2(t+1)^2}$, $x = \frac{t}{2} + \frac{1}{2(t+1)} - \frac{1}{2}$, $x(\frac{1}{2}) = \frac{1}{12}$.
5	**(c)**	$3\mathbf{a} - 4\mathbf{b} = (6, -7, 12)$, length $\sqrt{229}$.
6	**(a)**	$y(2) = 8 + \frac{b}{8} = 1$, so $b = -56$.
7	**(b)**	$M_y = \frac{1}{x}$, $N_x = \frac{K}{x}$, so $K = 1$.
8	**(a)**	Bernoulli $n = 3$, $u = y^{-2}$: $u' + 2u = \frac{e^{-2x}}{x}$.
9	**(e)**	$\det = 1 - 5k \neq 0$, so $k \neq \frac{1}{5}$.
10	**(c)**	Highest derivative is $y^{(4)}$, so the order is 4.
11	**(c)**	$4a - 2b = 2$, $a - b = -2$ give $a = 3$, $b = 5$, so $a + b = 8$.
12	**(e)**	Row reduce: $x_3 = -3x_4$, $x_2 = 4x_4$, $x_1 = -6x_4$, so $\mathbf{u} = (-6, 4, -3, 1)$.
13	**(e)**	$\ln\lvert\sec y\rvert = \frac{1}{2}\ln(x^2+1) + C$, so $y = \sec^{-1}\left(c\sqrt{1+x^2}\right)$.
14	**(c)**	$\rho = x^{1/3}$: $(x^{1/3}y)' = 4x^{1/3}$, so $y = 3x + c\,x^{-1/3}$.
15	**(e)**	Only $e^x y' + (\ln x)y = \tan x$ has y, y' to the first power with coefficients in x.
16	**(d)**	Homogeneous linear equation: contains $\mathbf{0}$, closed under $+$ and scalars, a subspace.
17	**(d)**	$y = \sin^{-1} x$ (since $y(0) = 0$), so $y(\frac{1}{2}) = \frac{\pi}{6}$.
18	**(e)**	$e^{-20k} = \frac{3}{5}$, and $\frac{9}{25} = (\frac{3}{5})^2$, so $t = 40$.
19	**(c)**	$F = \frac{x^2}{y} + \frac{y^2}{x^3} + g(y)$, $g' = \frac{1}{\sqrt{y}}$, so $+2\sqrt{y} = c$.
20	**(d)**	$v = x + y$: $v' = \frac{v+1}{v}$, so $v - \ln\lvert v+1\rvert = x + C$, i.e. $y = \ln(x + y + 1) + c$.